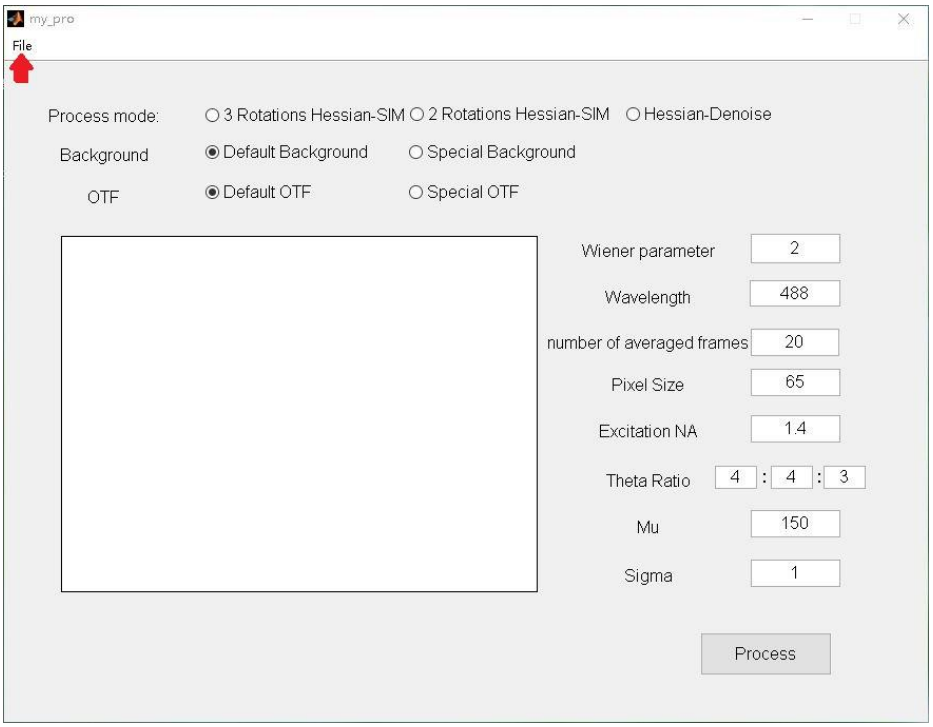
Manual for using Hessian-SIM software

If MATLAB software has been installed in the computer, please run the main.m in the code folder and turn to step③. Otherwise, follow the step-by-step guidance as below:

① If this is the first time of your executing the software, please install the 32-bit MCRInstaller.exe (R2012a version 7.17) on the computer with the administrator rights. The MCRInstaller can be downloaded from the following website: www.mathworks.com/products/compiler/matlab-runtime.html

② Run the 'Hessian_SIM.exe'.

③ Click the 'file' button, choose the raw data 'Actin_97hz_0.5ms exposure raw data' in the 'raw-data' folder or other TIF format data in the computer.

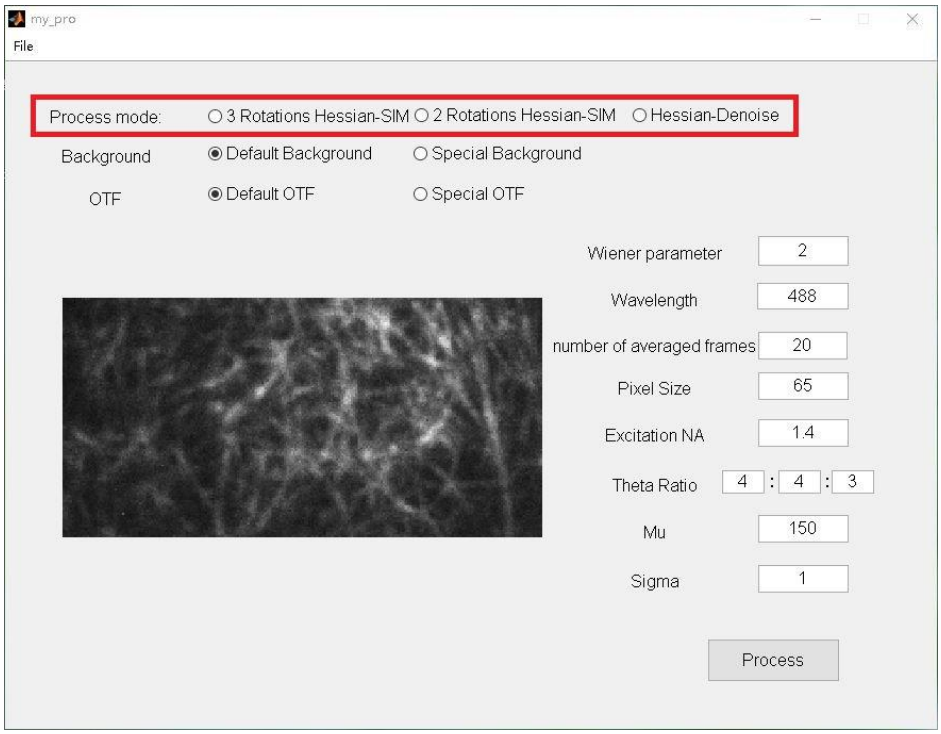


④ Choose the process mode:

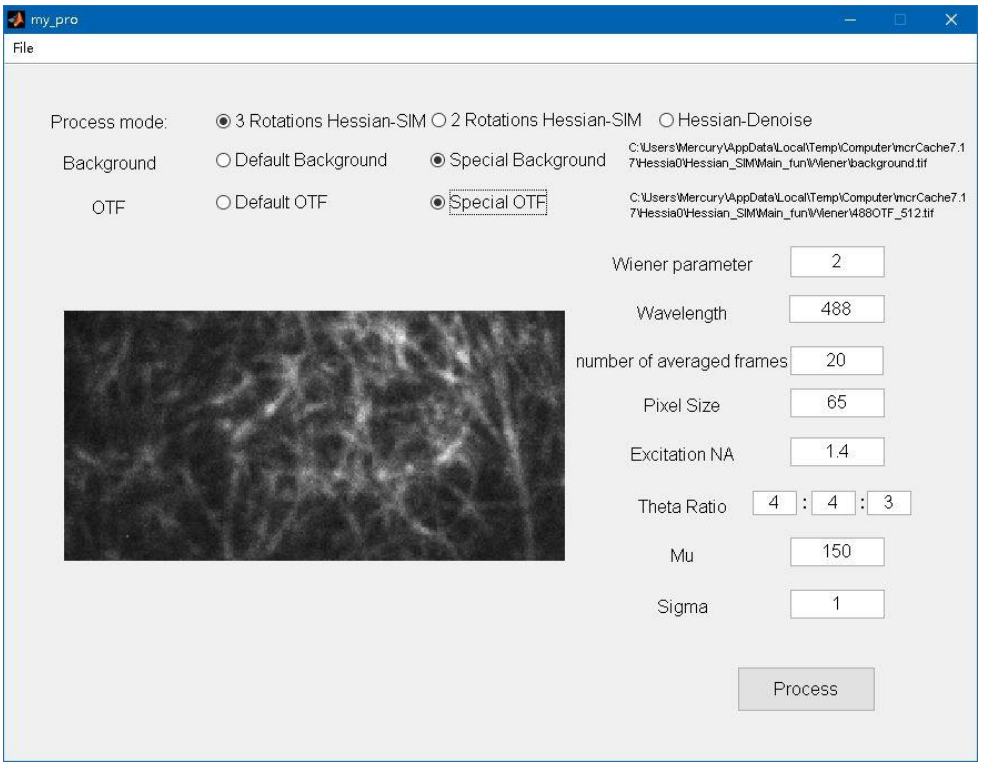
3 beam Hessian-SIM: We reconstruct the super resolution images with Wiener-SIM, Hessian-SIM, TV-SIM and Running-Average methods by the 3 beam raw data under this mode.

2 beam Hessian-SIM: We reconstruct the super resolution images with Wiener SIM, Hessian SIM, TV SIM and Running-Average methods by the 2 beam raw data under this mode (2-beam SIM refer to: Shroff SA, Fienup JR, Williams DR. Lateral superresolution using a posteriori phase shift estimation for a moving object: experimental results. ***J Opt Soc Am A*** Opt Image Sci Vis. 2010 Aug 1;27(8):1770-82.).

Hessian-Denoise: We denoise results obtained with the commercial SIM software with Hessian algorithm under this mode.



⑤ Choose the Background and OTF images:
Default: using the default Background and OTF images embedded in the program.
Special: using the Background and OTF images generated from other SIM systems.



⑥ Change the parameters:
Wiener parameter: the parameter of Wiener filter, depending on the signal to noise ratio (**Supplementary: Section 1**).

Wavelength: the excitation wavelength.
Number of averaged frames: the number of averaged raw images to improve the estimation accuracy of parameters (Supplementary: Section 2.2), which should be smaller than the one-ninth of total frames (*3 beam mode*) or one-sixth of total frames (*2 beam mode*) of raw SIM data.
Pixel Size: the pixel size of camera in sample plane, unit: nanometer.
Excitation NA: the excitation NA of illumination pattern.
Theta Ratio: the ratio of *Theta1*, *Theta2*, and *Theta3*, which depends on the pattern displayed on the SLM. For example, if the period is 5.5 pixel, we moved 2 pixels each time in one period, so the ratio is 2:2:1.5 in our system. In commercial SIM system, the theta moves 120 degree each time, the *Theta Ratio* should be 1:1:1.
Mu, Sigma: the parameters of Hessian SIM (**Supplementary Section 2.4**)
Parameter list:

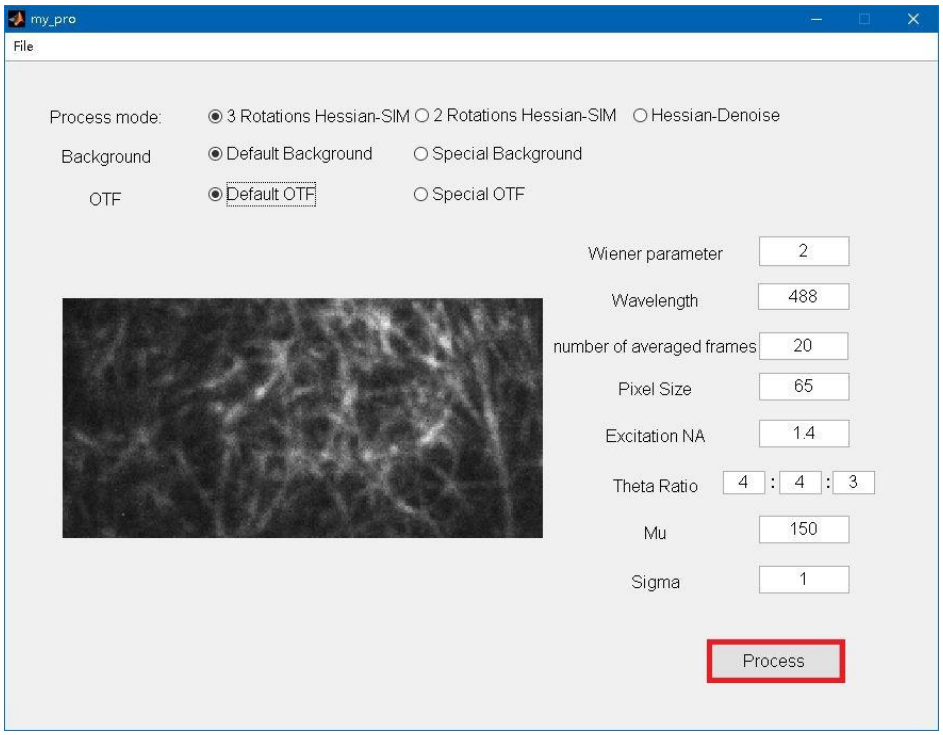
	A	*B*	*C*	*D*
Process mode	3 beam Hessian-SIM	2 beam Hessian-SIM	Hessian denoise	Hessian denoise
Background	default	default	/	/
OTF	default	default	/	/
Wiener parameter	2	1	/	/
Wavelength	488	488	/	/
Number of averaged frames	⩽20	⩽12	/	/
Pixel Size	65	65	/	/
Excitation NA	1.4	0.9	/	/
Theta Ratio	4:4:3	1:1:1	/	/
Mu	150	100	150	150
Sigma	1	1	1	0.8

Note: *A* is the HUVEC labeled with Lifeact-EGFP (Actin_97hz_0.5ms exposure raw data, Actin_97hz_7ms exposure raw data), *B* is the 2-beam 2D simulation data, *C* is the 3D N-SIM images of HEK293 cell labeled with Lifeact-EGFP, *D* is the 2D N-SIM images of HUVEC labeled with Lifeact-EGFP.

⑦ Click the 'Process' button. The program will reconstruct the super resolution images with Wiener SIM, Hessian SIM, TV SIM and Running-Average methods (*3 beam or 2 beam Hessian SIM mode*), or be applied to the commercial SIM super-resolution image with Hessian de-noise method that does not require knowledge of the system background and OTF (*Hessian de-noise mode*).

Reconstruction process

The Wait bar Window will display the process rate of reconstruction.



Result

The results are presented in the following folders:

'submit\raw-data\Pseudo-TIRF\TIRF_xxx.tif' is the averaged pseudo TIRF images with each nine raw data;

'submit\raw-data\SIM-Wiener\re-xxx.tif' is the Wiener reconstruction result;

'submit\raw-data\SIM-Hessian\Hessian_xxx.tif' is the Hessian reconstruction result;

'submit\raw-data\SIM-Hessian\Hessian-Denoise-xxx.tif' is the Hessian denoising result of super resolution image;

'submit\raw-data\SIM-TV\TV_xxx.tif' is the TV reconstruction result;

'submit\raw-data\Running-Average\RunningAverage_xxx.tif' is the Running Average of Wiener reconstruction;

Note

1) Due to the method of computing Hessian penalty in our algorithm, the result of the first and the last two frames of the Hessian SIM still contains artifacts, which are usually discarded. Similarly, if the frame number of SR image is less than 3, the *t* and *z* axes of Hessian SIM will be turned off. This means that σ will be set to 0 regardless of the number inputted in the menu.
2) The default OTF and background embedded in the software only match our home-built TIRF-SIM system. Upon reconstructing raw data from other SIM systems, the specific OTF and background images should be provided before the reconstruction process. The background was generated by the camera of the system, and its size cannot be smaller than the size of raw SIM images. The special wavelength specific OTF could be generated by the SIM system with the protocol in the *Nature Protocols 12, 988–1010 (2017)*.
3) The 64-bit Operating System is preferred to avoid the interrupt of out of memory.